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Lecture 5 Lab
Local Serverless Computing and Event-Driven Image Processing Pipeline
https://github.com/youssef-eldesouky/Cloud-and-Mobile-Computing

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1. Project Structure:

```
Lecture5-local-serverless-lab
│   docker-compose.yml
│
├────data
│   ├────input
│   └────output
├────event_source
│       Dockerfile
│       requirements.txt
│       watcher.py
│
├────functions
│   ├────image_resizer
│   │       app.py
│   │       Dockerfile
│   │       requirements.txt
│   │
│   └────notifier
│           app.py
│           Dockerfile
│           requirements.txt
│
└────router
        Dockerfile
        event_router.py
        requirements.txt
```

## 2. Run the Full Local Pipeline:

```
docker compose up --build
```

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose up --build
[+] up 11/11
 ✓ Image redis:7-alpine Pulled
[+] Building 27.9s (37/37) FINISHED
 [+] up 21/21outer] resolving provenance for metadata file                                                    0.0s
  ✓ Image redis:7-alpine                                Pulled                                                 8.6s
  ✓ Image lecture5-local-serverless-lab-event-router    Built                                                  28.8s
  ✓ Image lecture5-local-serverless-lab-event-source    Built                                                  28.8s
  ✓ Image lecture5-local-serverless-lab-image-resizer   Built                                                  28.8s
  ✓ Image lecture5-local-serverless-lab-notifier        Built                                                  28.8s
  ✓ Network lecture5-local-serverless-lab_default       Created                                                0.1s
  ✓ Container lab5-redis                                Created                                                0.3s
  ✓ Container lab5-image-resizer                        Created                                                0.3s
  ✓ Container lab5-notifier                             Created                                                0.3s
  ✓ Container lab5-event-source                         Created                                                0.2s
  ✓ Container lab5-event-router                         Created                                                0.2s
Attaching to lab5-event-router, lab5-event-source, lab5-image-resizer, lab5-notifier, lab5-redis
lab5-redis   | 1:C 03 May 2026 20:13:13.534 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
lab5-redis   | 1:C 03 May 2026 20:13:13.534 * Redis version=7.4.8, bits=64, commit=00000000, modified=0, pid=1, just started
lab5-redis   | 1:C 03 May 2026 20:13:13.534 # Warning: no config file specified, using the default config. In order to specify a config file use redis-server /path/to/redis.conf
lab5-redis   | 1:M 03 May 2026 20:13:13.534 * monotonic clock: POSIX clock_gettime
lab5-redis   | 1:M 03 May 2026 20:13:13.535 * Running mode=standalone, port=6379.
lab5-redis   | 1:M 03 May 2026 20:13:13.536 * Server initialized
lab5-redis   | 1:M 03 May 2026 20:13:13.536 * Ready to accept connections tcp
lab5-image-resizer  |  * Serving Flask app 'app'
lab5-image-resizer  |  * Debug mode: off
lab5-image-resizer  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lab5-image-resizer  |  * Running on all addresses (0.0.0.0)
lab5-image-resizer  |  * Running on http://127.0.0.1:5000
lab5-image-resizer  |  * Running on http://172.19.0.2:5000
lab5-image-resizer  | Press CTRL+C to quit
lab5-notifier       |  * Serving Flask app 'app'
lab5-notifier       |  * Debug mode: off
lab5-notifier       | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lab5-notifier       |  * Running on all addresses (0.0.0.0)
lab5-notifier       |  * Running on http://127.0.0.1:5000
lab5-notifier       |  * Running on http://172.19.0.3:5000
lab5-notifier       | Press CTRL+C to quit
lab5-event-router   |   File "/app/event_router.py", line 42
lab5-event-router   |     print(f"
lab5-event-router   |           ^
lab5-event-router   | SyntaxError: unterminated f-string literal (detected at line 42)
lab5-event-router exited with code 1
lab5-event-source   | Event Source started. Watching folder: /data/input
```

```
docker compose ps
```

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose ps
NAME                IMAGE                                   COMMAND                  SERVICE         CREATED          STATUS          PORTS
lab5-event-source   lecture5-local-serverless-lab-event-source   "python watcher.py"      event-source    23 minutes ago   Up 23 minutes
lab5-image-resizer  lecture5-local-serverless-lab-image-resizer  "python app.py"          image-resizer   23 minutes ago   Up 23 minutes   0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp
lab5-notifier       lecture5-local-serverless-lab-notifier       "python app.py"          notifier        23 minutes ago   Up 23 minutes   0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp
lab5-redis          redis:7-alpine                          "docker-entrypoint.s…"   redis           23 minutes ago   Up 23 minutes   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose ps
NAME                IMAGE                                   COMMAND                  SERVICE         CREATED          STATUS          PORTS
lab5-event-source   lecture5-local-serverless-lab-event-source   "python watcher.py"      event-source    25 minutes ago   Up 25 minutes
lab5-image-resizer  lecture5-local-serverless-lab-image-resizer  "python app.py"          image-resizer   25 minutes ago   Up 25 minutes   0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp
lab5-notifier       lecture5-local-serverless-lab-notifier       "python app.py"          notifier        25 minutes ago   Up 25 minutes   0.0.0.0:5002->5000/tcp, [::]:5002->5000/tcp
lab5-redis          redis:7-alpine                          "docker-entrypoint.s…"   redis           25 minutes ago   Up 25 minutes   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
```

The terminal shows that the services are running and that the event source is watching the input folder

3. Creating Test Image Locally:

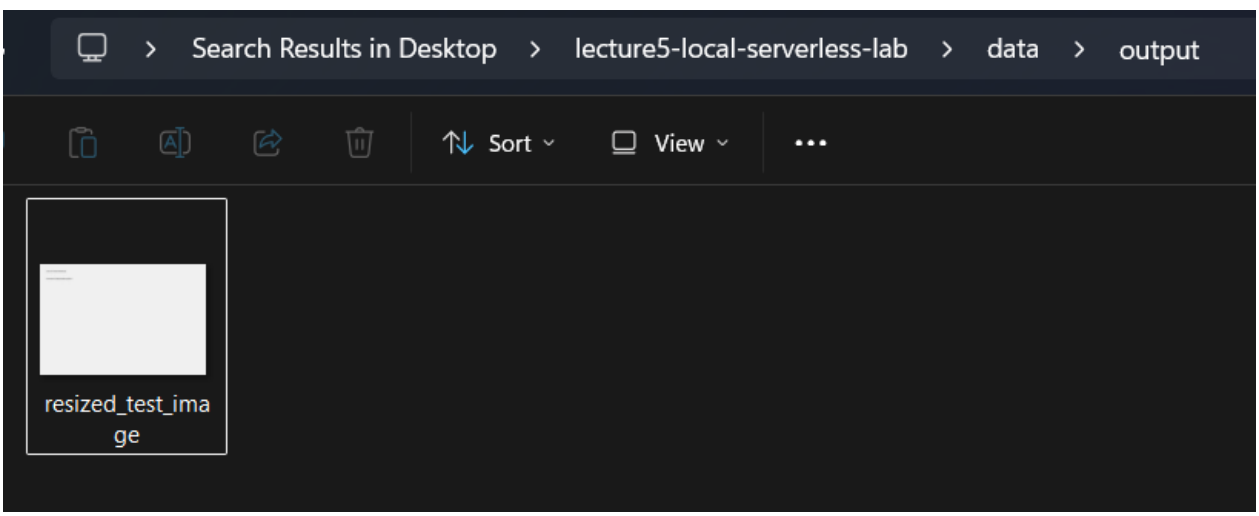
```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose exec -T image-resizer python - <<'PY'
from PIL import Image, ImageDraw

img = Image.new("RGB", (1200, 800), (240, 240, 240))
draw = ImageDraw.Draw(img)

draw.text((50, 50), "Lecture 5 Local Serverless Lab", fill=(0, 0, 0))
draw.text((50, 100), "Event-driven image processing pipeline", fill=(0, 0, 0))

img.save("/data/output/test_image.png")

print("Created /data/output/test_image.png")
PY
Created /data/output/test_image.png
```

The event-source container detected test_image.png and published an event to Redis

4. Checking the Output:

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ ls data/input
test_image.png
```

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose logs event-source
docker compose logs event-router
docker compose logs image-resizer
docker compose logs notifier
lab5-event-source   | Event Source started. Watching folder: /data/input
lab5-event-source   | Published event: {
lab5-event-source   |   "event_id": "03bc8a7b-f6d0-4f03-b45d-3dbc561d091d",
lab5-event-source   |   "event_type": "image.uploaded",
lab5-event-source   |   "file_name": "test_image.png",
lab5-event-source   |   "file_path": "/data/input/test_image.png",
lab5-event-source   |   "width": 300,
lab5-event-source   |   "created_at": "2026-05-03T20:46:51.805978"
lab5-event-source   | }
lab5-event-router   |   File "/app/event_router.py", line 42
lab5-event-router   |     print(f"
lab5-event-router   |           ^
lab5-event-router   | SyntaxError: unterminated f-string literal (detected at line 42)
lab5-image-resizer  |  * Serving Flask app 'app'
lab5-image-resizer  |  * Debug mode: off
lab5-image-resizer  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lab5-image-resizer  |  * Running on all addresses (0.0.0.0)
lab5-image-resizer  |  * Running on http://127.0.0.1:5000
lab5-image-resizer  |  * Running on http://172.19.0.2:5000
lab5-image-resizer  | Press CTRL+C to quit
lab5-image-resizer  | 172.19.0.1 - - [03/May/2026 20:57:41] "POST /resize HTTP/1.1" 400 -
lab5-image-resizer  | 172.19.0.1 - - [03/May/2026 20:59:30] "POST /resize HTTP/1.1" 200 -
lab5-image-resizer  | 172.19.0.1 - - [03/May/2026 21:09:35] "POST /resize HTTP/1.1" 200 -
lab5-image-resizer  |  * Serving Flask app 'app'
lab5-image-resizer  |  * Debug mode: off
lab5-image-resizer  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lab5-image-resizer  |  * Running on all addresses (0.0.0.0)
lab5-image-resizer  |  * Running on http://127.0.0.1:5000
lab5-image-resizer  |  * Running on http://172.19.0.2:5000
lab5-image-resizer  | Press CTRL+C to quit
lab5-image-resizer  | 172.19.0.1 - - [03/May/2026 21:09:56] "POST /resize HTTP/1.1" 200 -
lab5-notifier   |  * Serving Flask app 'app'
lab5-notifier   |  * Debug mode: off
lab5-notifier   | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
lab5-notifier   |  * Running on all addresses (0.0.0.0)
lab5-notifier   |  * Running on http://127.0.0.1:5000
lab5-notifier   |  * Running on http://172.19.0.3:5000
lab5-notifier   | Press CTRL+C to quit
```



5. Testing the Function Directly:

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ curl -X POST http://localhost:5001/resize   -H "Content-Type: application/json"   -d '{"file_path":"/data/i
nput/test_image.png","file_name":"test_image.png","width":200}'
{"function":"image-resizer","input_file":"test_image.png","new_size":[200,133],"original_size":[1200,800],"output_file":"/data/output/resized_test_image.png","processing_ms":110.43,"status":"succ
ess"}
```

youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ curl -X POST http://localhost:5001/resize  -H "Content-Type: application/json"  -d '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'

{"function":"imageresizer","input_file":"test_image.png","new_size":[200,133],"original_size":[1200,800],"output_file":"/data/output/resized_test_image.png","processing_ms":110.43,"status":"success"}

6. Cold Starting the Mini-Experiment:

```
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ curl -w "
Total time: %{time_total}s
" -X POST http://localhost:5001/resize   -H "Content-Type: application/json"   -d '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'
{"function":"image-resizer","input_file":"test_image.png","new_size":[200,133],"original_size":[1200,800],"output_file":"/data/output/resized_test_image.png","processing_ms":74.03,"status":"succe
ss"}

Total time: 0.117458s
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ docker compose restart image-resizer
[+] restart 0/1
 ⠿ Container lab5-image-resizer Restarting                                                                                                                                                  2.0s
youssefeldesouky@LAPTOP-UTRLSR8U:/mnt/c/Users/PC/Desktop/lecture5-local-serverless-lab$ curl -w "
Total time: %{time_total}s
" -X POST http://localhost:5001/resize   -H "Content-Type: application/json"   -d '{"file_path":"/data/input/test_image.png","file_name":"test_image.png","width":200}'
{"function":"image-resizer","input_file":"test_image.png","new_size":[200,133],"original_size":[1200,800],"output_file":"/data/output/resized_test_image.png","processing_ms":54.85,"status":"succe
ss"}

Total time: 0.061990s
```

7. Answering the Cold-start Measurement Table:

| Measurement | Value |
|---|---|
| Warm request time | 0.117458s |
| First request after restart | 0.061990s |
| Difference | 0.055468 |
| Reason / explanation | the first request after a function container starts may be slower than later requests |

8. Answering Reflection Questions:

1. **What is the event source in this lab?**
   The event source is the watcher.py container. It watches the data/input folder and publishes an event when a new image appears.
2. **What is the event router?**
   The event router is the event_router.py container. It reads events from Redis Streams and sends them to the correct function endpoints.
3. **What are the event destinations?**
   The event destinations are the image-resizer function and the notifier function.
4. **Why is this pipeline loosely coupled?**
   Because each part works separately through events. The source does not directly depend on the functions, and the functions do not need to know who sent the event.
5. **What happened after restarting the image-resizer container?**
   The first request after restart was usually a little slower than the warm request because the container had to start again.
6. **How is this similar to serverless cold start?**
   It is similar because a function can be slower on the first request after startup, just like a real serverless function may take longer when it is not already running.
7. **What is missing compared with real cloud serverless platforms?**
   It does not have real autoscaling, managed event services, automatic scaling to zero, runtime isolation like cloud platforms, or built-in monitoring and deployment tools.
8. **How could this pipeline be extended to support image moderation, OCR, or ML enrichment?**
   More functions could be added to the router. For example, one function could check unsafe images, another could extract text using OCR, and another could run an ML model to classify the image.

9. Final Project Structure:

```
├── data
│   ├── input
│   │   └── test_image.png
│   └── output
│       └── resized_test_image.png
├── docker-compose.yml
├── event_source
│   ├── Dockerfile
│   ├── requirements.txt
│   └── watcher.py
├── functions
│   ├── image_resizer
│   │   ├── Dockerfile
│   │   ├── app.py
│   │   └── requirements.txt
│   └── notifier
│       ├── Dockerfile
│       ├── app.py
│       └── requirements.txt
└── router
    ├── Dockerfile
    ├── event_router.py
    └── requirements.txt
```